High-frequency data and limit order books

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Reminder from the last lecture

- Point processes, random counting measures, intensity measures.
- Poisson random measures:
  - definition ; superposition, marking, thinning ; some conditional distributions.
- Poisson point process on $[0, +\infty)$:
  - as special case of a Poisson random measure ;
  - via direct construction ; time change theorems.
- If N is a time-homogeneous Poisson process with intensity parameter λ, then the density of $T_n | T_1, \dots, T_{n-1}$ is
$$f_{T_n|T_1,\dots,T_{n-1}}(t) = \lambda e^{-\lambda(t-T_{n-1})}.$$
- If N is a Poisson process with intensity function $\lambda(t)$, the density of $T_n | T_1, \dots, T_{n-1}$ is
$$f_{T_n|T_1,\dots,T_{n-1}}(t) = \lambda(t) e^{-\int_{T_{n-1}}^{t} \lambda(u)\, du}.$$

Conditional intensity of a counting process – Intuition I

- Conditional intensity provides a causal description of a point process on $[0, +\infty)$.
- Let $f_n(t|t_1, \dots, t_{n-1})$, $t \geq t_{n-1}$, be the probability density function for T_n given $\{T_1 = t_1, \dots, T_{n-1} = t_{n-1}\}$. The survival function is:

$$S_n(t) = \mathbf{P}(T_n > t|t_1, \dots, t_{n-1}) = 1 - \int_{t_{n-1}}^{t} f_n(u|t_1, \dots, t_{n-1})\, du.$$

- Let us set $h_n(t|t_1, \dots, t_{n-1}) = \dfrac{f_n(t|t_1, \dots, t_{n-1})}{S_n(t|t_1, \dots, t_{n-1})}$. Then:

$$f_n(t|t_1, \dots, t_{n-1}) = h_n(t|t_1, \dots, t_{n-1}) e^{-\int_{t_{n-1}}^{t} h_n(u|t_1, \dots, t_{n-1})\, du}.$$

- Moreover, $\mathbf{E}[N(t_{n-1} + \epsilon) - N(t_{n-1})|t_1, \dots, t_{n-1}] \approx h_n(t_{n-1} + |t_1, \dots, t_{n-1})\, \epsilon, \quad (\epsilon \ll 1).$

Conditional intensity of a counting process – Intuition II

- Hence, if we define a function λ^* successively on a sample path $(0 < t_1 < \ldots < t_n < \ldots)$ by

$$\lambda^*(t) = \begin{cases} h_1(t), & 0 < t \leq t_1, \\ h_n(t|t_1, \ldots, t_{n-1}), & t_{n-1} < t \leq t_n, n \geq 2, \end{cases}$$

  then this function characterizes the probability structure of the point process, since f_n and S_n expressed in terms of λ^*:

$$f_n(t|t_1, \ldots, t_{n-1}) = \lambda^*(t) e^{-\int_{t_{n-1}}^{t} \lambda^*(u)\, du},$$

  and we may write the intuitive formula:

$$\mathbf{E}[dN_t| \text{ history up to time } t-] \approx \lambda^*(t)\, dt.$$

- λ^* is called the conditional intensity of the point process.

Conditional intensity of a counting process I

- General theory relies on the Doob-Meyer decomposition.
- Let $(N_t)_{t\geq 0}$ be an integrable counting process with generated filtration $(\mathcal{F}_t)_{t\geq 0}$. By the Doob-Meyer decomposition of a submartingale, there exists a unique increasing predictable process A, with $A_0 = 0$ a.s., such that $M_t = N_t - A_t$ is a martingale.
- If we assume further that $A_t = \int_0^t \lambda_s \, ds$ for some predictable process $(\lambda_s)_{s\geq 0}$, then $N_t - \int_0^t \lambda_s \, ds$ is a martingale.

Definition (Intensity of a counting process)

A (predictable) process $(\lambda_s)_{s\geq 0}$ is called the (predictable) intensity of a counting process $(N_t)_{t\geq 0}$ if $M_t = N_t - \int_0^t \lambda_s \, ds$ is a martingale.

- A is called a compensator of the counting process, and the martingale is called M the compensated counting process.

Conditional intensity of a counting process II

- For any non-negative predictable process h, we have

$$\mathbf{E}\left[\int_0^\infty h_t \, dN_t\right] = \mathbf{E}\left[\int_0^\infty h_t \lambda_t \, dt\right].$$

- For any $t, \epsilon \geq 0$, we thus have $\mathbf{E}[N_{t+\epsilon} - N_t \mid \mathcal{F}_t] = \mathbf{E}\left[\int_t^{t+\epsilon} \lambda_s \, ds \mid \mathcal{F}_t\right]$.
- Interpretation:

$$\lambda_{t_+}|\mathcal{F}_t = \lim_{\epsilon \to 0} \frac{1}{\epsilon} \mathbf{E}[N_{t+\epsilon} - N_t | \mathcal{F}_t].$$

- Condensed notation:

$$\mathbf{P}[dN_t = 1|\mathcal{F}_{t-}] = \mathbf{E}[dN_t|\mathcal{F}_{t-}] = \lambda_t \, dt.$$

- If N is a Poisson process, the conditional intensity is equal to the intensity function.

Log-likelihood of a counting process I

Proposition (Log-likelihood of a point process)

The log-likelihood of a counting process $(N_t)_{t\geq 0}$ with conditional intensity $(\lambda^*_t)_{t\geq 0}$ observed on $[0, T]$ is

$$\log \mathcal{L}_T = \int_0^T \log \lambda^*_s \, dN_s - \int_0^T \lambda^*_s \, ds.$$

Sketch of the proof [10, 3]: We write a likelihood function with the conditional intensity via successive conditioning. Let $0 < t_1 < \ldots < t_n < T$ be an observation of the counting process N on $[0, T]$. A likelihood function is

$$\mathcal{L}_T = f_1(t_1)\, f_2(t_2|t_1) \ldots f_n(t_n|t_1, \ldots, t_{n-1})\, S_{n+1}(T|t_1, \ldots, t_n).$$

Log-likelihood of a counting process II

Given our construction of the conditional intensity, setting $t_0 = 0$, we have

$$\mathcal{L}_T = \prod_{i=1}^{n} \left[\lambda^*(t_i) \exp\left(- \int_{t_{i-1}}^{t_i} \lambda^*(u)\, du \right) \right] \times \exp\left(- \int_{t_n}^{T} \lambda^*(u)\, du \right),$$

and taking the logarithm: $\log \mathcal{L}_T = \sum_{i=1}^{n} \log \lambda^*(t_i) - \int_0^T \lambda^*(u)\, du$.

- *Example:* In the case of a time-homogeneous Poisson process, for an observed sample $0 < t_1 < \ldots < t_n < T$, the log-likelihood is:

$$\log \mathcal{L}_T = n \log \lambda - \lambda T.$$

The maximum-likelihood estimator of the intensity is thus $\hat{\lambda}_{MLE} = \frac{n}{T}$.

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One-dimensional Hawkes processes

Definition (Linear self-exciting process)

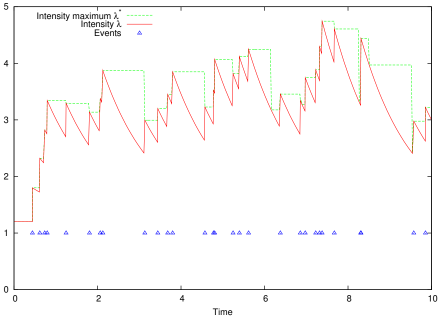
A counting process N is called a (linear, one-dimensional) Hawkes process if it is a counting process with conditional intensity:

$$\lambda_t = \mu(t) + \int_0^t \nu(t-s)dN_s = \mu(t) + \sum_{0<t_i<t} \nu(t-t_i),$$

where $\mu : \mathbb{R} \mapsto \mathbb{R}_+$ and $\nu : \mathbb{R}_+ \mapsto \mathbb{R}_+$ are deterministic functions.

- ν expresses the positive influence of the past events t_i on the current value of the intensity process. Hawkes processes are self-exciting processes.
- μ is the baseline intensity ($\lambda_t \geq \mu(t)$).
- Original paper by [5]. Used in geology (earthquakes replicas), finance (see later in the course), biology (neuron excitation, multidimensional case, see below).

Sample path, kernels



Sample path of a 1D-Hawkes process with a simple exponential kernel and parameters $\mu = 1.2, \alpha_1 = 0.6, \beta_1 = 0.8$.

- Basic definitions often use a simple exponential kernel:

$$\nu(t) = \alpha e^{-\beta t} \mathbf{1}_{\mathbb{R}_+}(t).$$

- Exponential kernels allow for efficient computations, especially for maximum-likelihood estimation (*see later*), but do not exhibit long-memory properties.
- P-exponential kernel:

$$\nu(t) = \sum_{j=1}^{P} \alpha_j e^{-\beta_j t} \mathbf{1}_{\mathbb{R}_+}(t).$$

Average intensity in the case of an exponential kernel

Proposition (Average intensity – exponential kernel case)

If $\lambda(t)$ *is the conditional intensity of a Hawkes process with constant baseline intensity* $\mu > 0$ *and exponential kernel* $\nu(t) = \alpha e^{-\beta t}$, $0 < \alpha < \beta$, *then*

$$\mathbf{E}[\lambda(t)] = \frac{\mu}{1 - \frac{\alpha}{\beta}} \left(1 - \frac{\alpha}{\beta} e^{-(\beta - \alpha)t} \right)$$

- Asymptotically, $\mathbf{E}[\lambda(t)] \sim \frac{\mu}{1 - \frac{\alpha}{\beta}}$, stable/stationary regime.
- The average number of points of a Hawkes process with exponential kernel on a time interval of length Δ is thus in a stationary regime $\frac{\mu \Delta}{1 - \frac{\alpha}{\beta}} = \frac{\mu \Delta}{1 - \|\nu\|_1}$.

Stability of a Hawkes process

Proposition (Average intensity in an asymptotic regime – generic case)

If a Hawkes process with constant baseline intensity μ and kernel ν has reached a stationary regime, the average intensity value is:

$$\lambda_\infty = \frac{\mu}{1 - \int_0^\infty \nu(x)dx}.$$

- If $\|\nu\|_1 < 1$, the Hawkes process is sub-critical / stationary. If $\|\nu\|_1 = 1$ (resp. > 1), the process is called critical (resp. supercritical) (*See discussion in finance later in the course*).
- In the case of exponential kernels, the stability condition is written $\sum_{j=1}^{P} \frac{\alpha_j}{\beta_j} < 1$.

Thinning for stochastic intensities

Theorem (Thinning)

Consider point process N with intensity λ_t on an interval $(0, T]$. Suppose we can find a process λ^*_t such that $\lambda_t \leq \lambda^*_t$ ($\mathbf{P}$-a.s.). Let $t^*_1, t^*_2, \dots t^*_n \in (0, T]$ be the points of the process N^* with intensity λ^*. For each of the points, attach a mark $p = 1$ with probability $\frac{\lambda(t^*_j)}{\lambda^*(t^*_j)}$, else $p = 0$. Then the points with marks $p = 1$ form a point process which is the same as N.

- Extension of the thinning method for Poisson processes [7] to point processes with stochastic intensities [8] .
- If λ^*_t can be constructed pathwise as a piecewise constant process, we can thus simulate the point process N with simple exponential random variables.

Simulation algorithm for a one-dimensional Hawkes process

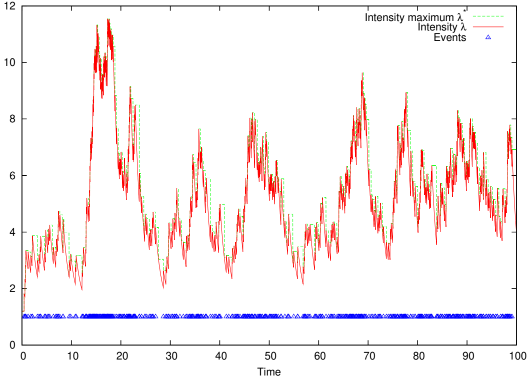
- Thinning algorithm for a Hawkes process with baseline μ and kernel ν:

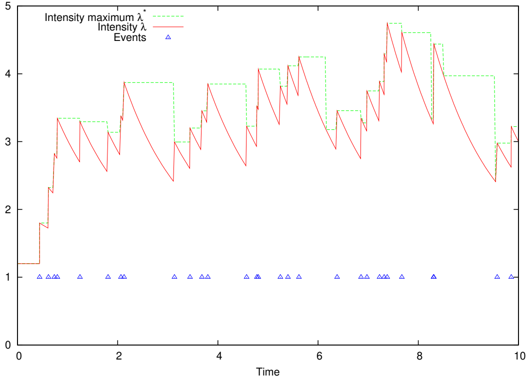
1. Initialization : Set $t \leftarrow 0$, $\lambda^* \leftarrow \mu(0)$ and an empty list of events.
2. Main loop : **While** $t < T$:
   (a) New event candidate : Generate $U \sim \mathcal{E}(\lambda^*)$ and set $t \leftarrow t + U$.
   If $t \geq T$, **Then** return the list of events.
   (b) Thinning : Generate $D \sim \mathcal{U}_{[0,1]}$.
   If $D \leq \frac{\lambda(t)}{\lambda^*}$,
   Then New event: add t to the list of events and update the maximum intensity $\lambda^* \leftarrow \lambda(t) + \nu(0)$;
   Else No new event: update the maximum intensity $\lambda^* \leftarrow \lambda(t)$.
3. Output: return the list of events.

$\lambda(t)$: it is crucial to observe that exponential kernels lead to an easy update of the jump part of the intensity, without having to loop on all previous events.

Path of a Hawkes process and its intensity

- Simulation of a one-dimensional Hawkes process with a single exponential kernel (parameters $P = 1, \mu = 1.2, \alpha_1 = 0.6, \beta_1 = 0.8$).





Log-likelihood of a Hawkes process I

- The log-likelihood of an observation $0 < t_1 < \ldots < t_n < T$ of a Hawkes process with baseline intensity $\mu(t)$ and kernel $\nu(t)$ is:

$$\log \mathcal{L}_T = \sum_{i=1}^{n} \log \left[\mu(t_i) + \sum_{k=1}^{i-1} \nu(t_i - t_k) \right] - \int_0^T \left[\mu(t) + \sum_{t_k < t} \nu(t - t_k) \right] dt.$$

- In general, computational cost in $\mathcal{O}(n^2)$.

Log-likelihood of a Hawkes process with a P-exponential kernel I

- In the case of a Hawkes process with a P-exponential kernel, the second term of the log-likelihood becomes

$$\Lambda(0,T) := \int_0^T \left[\mu(t) + \sum_{t_k<t} \nu(t-t_k)\right] dt = \int_0^T \mu(t)dt + \sum_{i=1}^{n}\sum_{j=1}^{P} \frac{\alpha_j}{\beta_j}\left(1 - e^{-\beta_j(T-t_i)}\right),$$

- and the double sum in the first term can be bypassed by a recursive computation ([8]):

$$\sum_{i=1}^{n} \log\left[\mu(t_i) + \sum_{j=1}^{P}\sum_{k=1}^{i-1} \alpha_j e^{-\beta_j(t_i-t_k)}\right] = \sum_{i=1}^{n} \log\left[\mu(t_i) + \sum_{j=1}^{P} \alpha_j R_j(i)\right],$$

where $R_j(1) = 0$ and $R_j(i) = e^{-\beta_j(t_i - t_{i-1})}(1 + R_j(i-1))$, for all $j = 1, \ldots, P$.

Log-likelihood of a Hawkes process with a P-exponential kernel II

Proposition (Log-likelihood of a Hawkes process, P-exponential kernel)

The log-likelihood of a Hawkes process $(N_t)_{t\geq 0}$ with a P-exponential kernel observed on $[0, T]$ is

$$\log \mathcal{L}_T = -\int_0^T \mu(s)ds - \sum_{i=1}^{n}\sum_{j=1}^{P}\frac{\alpha_j}{\beta_j}\left(1 - e^{-\beta_j(T-t_i)}\right) + \sum_{i=1}^{n}\log\left[\mu(t_i) + \sum_{j=1}^{P}\alpha_j R_j(i)\right].$$

- Maximum-likelihood estimator numerically available.
- Analytical gradients available for better optimization.
- Computational cost in $\mathcal{O}(n)$. In the case of non-exponential kernel, the absence of recursive evaluations may lead to prohibitive computational times.

Maximum-likelihood estimation

- [9]: The MLE estimator $\hat{\theta}_T = (\hat{\mu}, (\hat{\alpha}_j)_{j=1,\ldots,P}, (\hat{\beta}_j)_{j=1,\ldots,P})$ is
  - *consistent*, i.e. converges in probability to the true values $\theta^* = (\mu, (\alpha_j)_{j=1,\ldots,P}, (\beta_j)_{j=1,\ldots,P})$ as $T \to \infty$:
  $$\forall \epsilon > 0, \quad \lim_{T\to\infty} P[|\hat{\theta}_T - \theta^*| > \epsilon] = 0.$$
  - *asymptotically normal*, i.e.
  $$\sqrt{T}\left(\hat{\theta}_T - \theta^*\right) \to \mathcal{N}(0, I^{-1}(\theta^*))$$
  where $I(\theta) = \left(\mathrm{E}\left[\frac{1}{\lambda}\frac{\partial\lambda}{\partial\theta_i}\frac{\partial\lambda}{\partial\theta_j}\right]\right)_{i,j}$.
  - *asymptotically efficient*, i.e. asymptotically reaches the lower bound of the variance of an estimate.

Change of time

- The "change of time" theorem proved for the non-homogeneous Poisson process can be extended in the case of a stochastic intensity [2].

Theorem (Change of time of a point process)

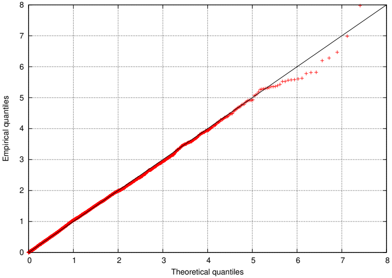
Let $(\tilde{N}_t)_{t\geq 0}$ be a point process with intensity $(\lambda_t)_{t\geq 0}$. Define for each t the stopping time $\tau(t)$ such that

$$\int_0^{\tau(t)} \lambda_s \, ds = t$$

Then $N_t = \tilde{N}_{\tau(t)}$ is a time-homogeneous Poisson process with intensity 1.

- Consequence: If the $\tilde{t}_i$'s are the points of $\tilde{N}$, then the quantities $\int_{\tilde{t}_{i-1}}^{\tilde{t}_i} \lambda_s \, ds$ are exponentially distributed with parameter 1.

Visual goodness-of-fit for a Hawkes process with a P-exponential kernel



- For a one-dimensional Hawkes process with a P-exponential kernel:

$$\Lambda(t_{i-1}, t_i) = \int_{t_{i-1}}^{t_i} \lambda(s)ds$$

$$= \int_{t_{i-1}}^{t_i} \mu(s)ds + \sum_{j=1}^{P} \frac{\alpha_j}{\beta_j} \left(1 - e^{-\beta_j(t_i - t_{i-1})}\right) A_j(i-1),$$

where $A_j(1) = 1$ and $A_j(i) = 1 + e^{-\beta_j(t_i - t_{i-1})} A_j(i-1)$.

- Quantile plots available as a *visual* indication of goodness-of-fit

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Branching representation of a Hawkes process I

Proposition (Branching representation of a Hawkes process)

Let $T > 0$. Let $\mu : [0, \infty) \to \mathbb{R}_+$ a deterministic baseline function and $\nu : [0, \infty) \to \mathbb{R}_+$ a deterministic function. Consider the following procedure:

1. *Generate a non-homogeneous Poisson process $\{t_i^{(0)}\}$ with intensity μ on $[0, T]$.*
2. *For each $t_i^{(0)}$, generate a non-homogeneous Poisson process $\{t_i^{(1)}\}$ with deterministic intensity $t \mapsto \nu(t - t_i^{(0)})$ on $[t_i^{(0)}, T]$.*
3. *Repeat operation 2. for each point of all the sets $\{t_i^{(1)}\}$, then for each point of all the sets $\{t_i^{(2)}\}$, etc. until there is no new point generated in $[0, T]$.*

Then the ordered set of all the points generated in $[0, T]$ is a Hawkes process N on $[0, T]$ with intensity $\lambda(t) = \mu(t) + \int_0^t \nu(t - s)\, dN_s$.

Branching representation of a Hawkes process II

- The average number of points (offspring/children) generated by one point (parent) is $\|\nu\|_1$. We retrieve the three regimes (sub-critical, critical, supercritical) discussed above depending on the value of $\|\nu\|_1$ compared to 1.
- Provides a new algorithm for the simulation of Hawkes processes.
- Provides a new algorithm for the estimation of Hawkes processes (see next slides).

The EM Algorithm – Generic case I

- Designed for likelihood estimation when some variables are latent (unobserved, missing).
- Let X be the observed data, Z the latent data, and θ the parameters. A complete (joint) likelihood function is

$$\ell(\theta; X, Z) = p(X, Z|\theta) = p(X|Z, \theta)p(Z|\theta),$$

  and

$$\ell(\theta; X) = p(X|\theta) = \int p(X|Z, \theta)p(Z|\theta)dZ$$

  is a marginal likelihood function.
- Direct maximization of the likelihood not possible.

The EM Algorithm – Generic case II

- EM algorithm principle (iterative): At iteration k, with parameter estimates $\theta^{(k)}$,
  - E-step (Expectation): compute the expected complete-data log-likelihood under the current distribution estimate $p(Z|X,\theta^{(k)})$:

$$Q(\theta|\theta^{(k)}) = \mathbf{E}_{Z\sim p(X,\theta^{(k)})}\left[\log p(X,Z|\theta)\right] = \int p(X|Z,\theta)p(Z|X,\theta^{(k)})dZ.$$

  - M-step (Maximization): maximize this function to update $\theta^{(k)}$:

$$\theta^{(k+1)} = \arg\max_{\theta} Q(\theta|\theta^{(k)}).$$

- Convergence guaranteed (but possibly to a local optimum only).

An EM algorithm for Hawkes processes I

- We can use the branching structure of the Hawkes process as a latent information: in the branching representation, each event is either a parent/immigrant or the child of a previous event.
- Let $(T_i)_{1 \leq i \leq n}$ be a sample data set of a Hawkes process. Let us define the latent variable

$$Z_{ij} = \begin{cases} 1 & \text{if } i = j \text{ and } T_i \text{ is a an immigrant,} \\ 1 & \text{if } T_i \text{ is a child of } T_j, \\ 0 & \text{otherwise.} \end{cases}$$

- A complete log-likelihood function is thus

$$\mathcal{L}_T = \sum_{i=1}^{n} Z_{ii} \log(\mu(t_i)) - \int_0^T \mu(t)\,dt + \sum_{i<j} \sum Z_{ij} \log(\nu(t_i - t_j)) - \sum_{j=1}^{n} \int_{t_j}^{T} \nu(t - t_j)\,dt.$$

An EM algorithm for Hawkes processes II

- We apply an EM algorithm with latent data Z. At iteration k, using the parameters $\theta^{(k)}$ (i.e. the distributions $\mu^{(k)}$ and $\nu^{(k)}$), the distribution estimate for Z is

$$p_{ij}^{(k)} := \mathbf{P}(Z_{ij}^{(k)} = 1 \mid (t_l), \theta^{(k)}) = \mathbf{E}\left[Z_{ij}^{(k)} \mid (t_l), \theta^{(k)}\right] = \begin{cases} \frac{\nu^{(k)}(t_i - t_j)}{\mu^{(k)}(t_i) + \sum_{l<i} \nu^{(k)}(t_i - t_l)} & \text{if } j < i, \\ \frac{\mu^{(k)}(t_i)}{\mu^{(k)}(t_i) + \sum_{l<i} \nu^{(k)}(t_i - t_l)} & \text{if } i = j, \\ 0 & \text{otherwise} \end{cases}$$

- Using this distribution we compute the expected complete log-likelihood $\mathbf{E}_{Z_{ij}^{(k)} \sim p_{ij}^{(k)}}[\mathcal{L}_T]$ (E-step) and then maximize the result w.r.t θ (M-step). Difficulty of the M-step depends on the parametric forms of μ and ν.

An EM algorithm for Hawkes processes III

- In the case of a Hawkes process with constant baseline intensity μ and single exponential kernel $\nu(t) = \alpha\beta e^{-\beta t}$, one can derive explicit (approximate) update equations:

$$\mu^{(k+1)} = \frac{1}{T}\sum_{i=1}^{n} p_{ii}^{(k)}, \quad \alpha^{(k+1)} = \frac{\sum_{i>j} p_{ij}^{(k)}}{n - \sum_{j=1}^{n} e^{-\beta^{(k)}(T-t_j)}},$$

$$\beta^{(k+1)} = \frac{\sum_{i>j} p_{ij}^{(k)}}{\sum_{i>j} p_{ij}^{(k)}(t_i - t_j) + \alpha^{(k+1)} \sum_{j=1}^{n}(T - t_j)e^{-\beta^{(k)}(T-t_j)}}.$$

(Approximation by using $\beta^{(k)}$ to compute $\alpha^{(k+1)}$ and $\beta^{(k+1)}$ instead of solving a system of non-linear equations).

An EM algorithm for Hawkes processes IV

- In summary:

EM for a Hawkes process with exponential kernel

1. Initialization $\theta^{(0)} = (\mu^{(0)}, \alpha^{(0)}, \beta^{(0)})$.
2. E-step: compute the probabilities $p_{ij}^{(k)}$.
3. M-step: update $\mu^{(k+1)}, \alpha^{(k+1)}, \beta^{(k+1)}$.
4. Repeat until $\|\theta^{(k+1)} - \theta^{(k)}\| < \epsilon$.

- See e.g. [6] for details and extensions.

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Multidimensional Hawkes processes

Definition (Multidimensional Hawkes processes)

A M-dimensional Hawkes process is a point process $\mathbf{N}_t = (N_t^1, \dots, N_t^M)$ with intensities $(\lambda_t^m)_{t\geq 0}$, $m = 1, \dots, M$ such that:

$$\lambda^m(t) = \lambda_0^m(t) + \sum_{n=1}^{M} \int_0^t g_{mn}(t-s)dN_s^n,$$

where $\mu : [0, \infty) \to \mathbb{R}_+$ and $g_{mn} : [0, \infty) \to \mathbb{R}_+$, $m, n = 1, \dots, M$ are deterministic functions.

- We may write using a vector notation: $\boldsymbol{\lambda}(t) = \boldsymbol{\mu} + \int_0^t \mathbf{G}(t-s) \cdot d\mathbf{N}_s$.
- In the case of exponential kernels, $\mathbf{G}(t) = \left(\alpha^{mn} e^{-\beta^{mn} t}\right)_{m,n=1,\dots,M}$.

Stability of multidimensional Hawkes processes

- If the process is in a stationary state, the average intensities $\boldsymbol{\mu}^t \in \mathbb{R}^M$should satisfy:

$$\boldsymbol{\mu} = \left(\mathbf{I} - \int_0^\infty \mathbf{G}(u) du \right)^{-1} \boldsymbol{\mu}$$

- If the spectral radius of the matrix $\boldsymbol{\Gamma} = \int_0^\infty \mathbf{G}(u) du$ is strictly smaller than 1, then the M-dimensional Hawkes process is stable.
- In the exponential kernel case, $\boldsymbol{\Gamma} = \left(\frac{\alpha^{mn}}{\beta^{mn}.} \right)_{m,n=1,\ldots,M}$

Simulation of multidimensional Hawkes processes by thinning

- The algorithm is formally identical to one in one dimension.
- Let $I_k(t) = \sum_{n=1}^{k} \lambda^n(t)$, $k = 1, \dots, M$.
- The total intensity is now $I_M(t)$, bounded above by I_M^*, and the next candidate event must not be generated with I_M^*.
- The thinning step is now:
  Generate $D \rightsquigarrow \mathcal{U}_{[0,1]}$.
  If $D \leq \frac{I_M(s)}{I_M^*}$,
  Then set s as a point of the process n_0, where n_0 is such that $\frac{I_{n_0-1}(s)}{I_M^*} < D \leq \frac{I_{n_0}(s)}{I_M^*}$, and go through the main loop again ;
  Else update $I_M^* \leftarrow I_M(s)$ and generate a new candidate point.

Log-likelihood of multidimensional Hawkes processes

- Log-likelihood of a multidimensional Hawkes process:

$$\log \mathcal{L}_T = \sum_{m=1}^{M} \log \mathcal{L}_T^m,$$

with: $\log \mathcal{L}_T^m = \int_0^T \log \lambda_s^m \, dN_s^m - \int_0^T \lambda_s^m \, ds.$

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Second-order characterization of Hawkes processes

Proposition (Second-order characterization of Hawkes processes [1])

A multivariate Hawkes process with stationary increments $N_t = (N_t^i)_{i=1,\dots,D}$ is uniquely defined by its first-order statistics (the expectation of the intensity) and its second-order statistics (the covariance structure), i.e. by

$$\begin{cases} \Lambda^i = \mathbf{E}[\lambda_t^i], \\ \mathrm{Cov}(dN_t^i, dN_{t'}^j) = \mathbf{E}(dN_t^i dN_{t'}^j) - \Lambda^i \Lambda^j \, dt \, dt'. \end{cases}$$

- More precisely, the kernel matrix $\varphi(t)$ of the Hawkes process is the unique solution of the matrix equation
$$g(t) = \varphi(t) + (\varphi \star g)(t), t \geq 0.$$
where the operator $\star$ stands for regular matrix multiplications where all the multiplications are replaced by convolutions, and the matrix $g(t)$ has components satisfying
$$g^{ij}(t)dt = \mathbf{E}(dN_t^i | dN_0 = 1) - \mathbf{1}_{\{i=j\}}\delta(t) - \Lambda^i dt.$$

Non-parametric estimation of Hawkes processes

- A characterization equation can also be developed for multidimensional marked Hawkes processes.
- The matrix convolution equation can be solved by numerical quadrature integration (original method proposed in [1]) or more recently via Deep Galerkin-like networks ([4], physics-informed networks).

Source : [4]

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Lab 3 - Questions I

Preliminary remarks:

- In this lab, you can code everything yourself. If you wish, you can also use this external Python library: `https://pypi.org/project/hawkes/`. If you would like to use another one, ask beforehand.
- Executing this lab may take a long time. Start by using small horizons when coding, and then run the notebook with larger horizons when you are satisfied with your preliminary results.
- You may write temporary results to files to avoid re-running simulations or estimations. To avoid any trouble during the evaluation execution, make sure that the names of your temporary result files start with the full name of your notebook. Temporary result files cannot be uploaded when submitting the lab.
- In you experiments, true values of the parameters should not be equal to one.

Lab 3 - Questions II

1. **Properties of Hawkes MLE estimates**. Consider simulations of a Hawkes process with constant baseline intensity μ and exponential kernel $\nu(t) = \alpha e^{-\beta t}$. Check that MLE estimators computed on simulated samples satisfy the expected statistical properties.
2. **Computational cost of Hawkes simulators**. Compare the computational cost of several simulation algorithms (and possibly several implementations of the same algorithm) and several types of kernels. Explain your experiments *in details*. Check that all your simulators are *correct*. Explain your results on the computational cost.
3. **MLE vs EM estimation.** Compare the MLE and the EM estimation a Hawkes process with constant baseline intensity μ and exponential kernel $\nu(t) = \alpha e^{-\beta t}$.

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